# ANALYSIS OF COUMARINS IN ESCIN SUBSTANCE

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**Abstract.** In the current process of globalization, the role of step-by-step quality control in quality assurance is incomparable. It is important to evaluate the amount of coumarins, which are foreign substances, in medicines obtained on the basis of horse chestnut. The work presents extensive materials on the analysis of coumarins in Escin substance. Quantitative analysis of coumarins was carried out by the HPLC-DAD method in gradient mode on Agilent 1260 (Agilent Technologies, Germany). The part of horse chestnut that stores the most coumarins is the bark, which contains 1.35±0.04% of coumarins. The primary raw materials for obtaining Escin substance are horse chestnut seeds, which contain 0.79±0.03% of coumarins. The amount of coumarins in "Escin substance" obtained by deep recleaning of seeds was found to be 0.4±0.02%. Thus, the introduction of the method of analysis of coumarins as a specific foreign substance into the regulatory documents of medicines obtained on the basis of horse chestnut will be an incentive to increase quality requirements by one step.

**Key words:** horse chestnut, escin, coumarin, esculin, fraxin, quality.

**Relevance.** The demand for the quality and harmlessness of drugs is increasing every year. It is important to develop methods for analyzing foreign substances contained in medicines and apply them to the production process. Coumarins as a foreign substance are found in the composition of medicines containing escin. One of the tasks that is waiting for a solution to develop their method of analysis and set the appropriate standards.

**Aim.** The purpose of the work is to analyze coumarins in Escin substance.

**Methods.** The bark and seeds of horse chestnut grown in Tashkent region, as well as Escin substance and the dry extract obtained from horse chestnut seeds were used. The study was carried out on the high-performance liquid chromatography (Agilent 1260 HPLC, Agilent Technologies, Germany), stationary phase Zorbax Eclipse XDB-C18 (size 150 × 3.0 mm, particle size 5.0 μm). The temperature of the stationary phase is 40°C. The mobile phase consisted of 0.1% formic acid (A) and acetonitrile (B) of variable composition at a flow rate of 1 ml/min. The gradient mode was optimized as follows: 0–3 min, 25–40% B; 3–10 min, 40–50% B; 10–15 min, 50% B; 15–18 min, 50–75% B; 18–22 min, 75–100% B; 22–25 min, 100–25% B. Injection volume – 50 μl. The detection wavelength is 330 nm.

The formula for calculating the amount of coumarins (%) in the tested samples:

$$X = \frac{S_1 \cdot a_0 \cdot V_1 \cdot P \cdot 100}{S_0 \cdot a_1 \cdot V_0 \cdot 100}$$

Where:

$S_0$ – peak area of the standard sample of esculin in chromatography, mAU
$S_1$ – peak area of coumarins in chromatography of the test solution samples, mAU
$a_0$ – weighted mass of standard sample of esculin, mg
$a_1$ - weighted mass of the test solution samples, mg
P- Degree of purity of the working standard sample of esculin, %
$V_0$ - diluted volume of the working standard sample of esculin, ml
$V_1$ - diluted volume of the test solution samples, ml

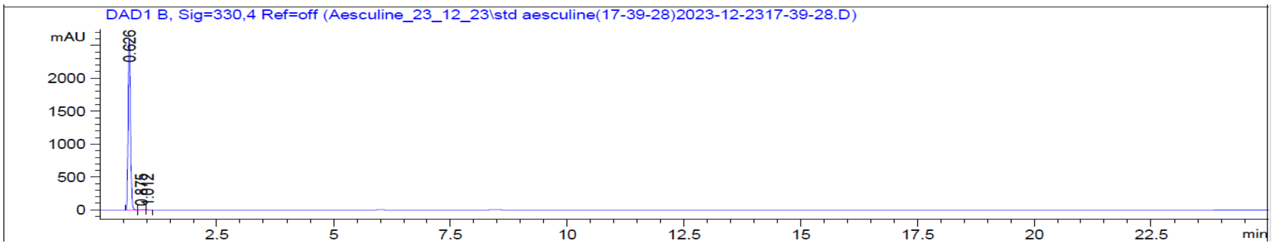


**Figure 1. High-performance liquid chromatography chromatogram of working standard solution of esculin.**

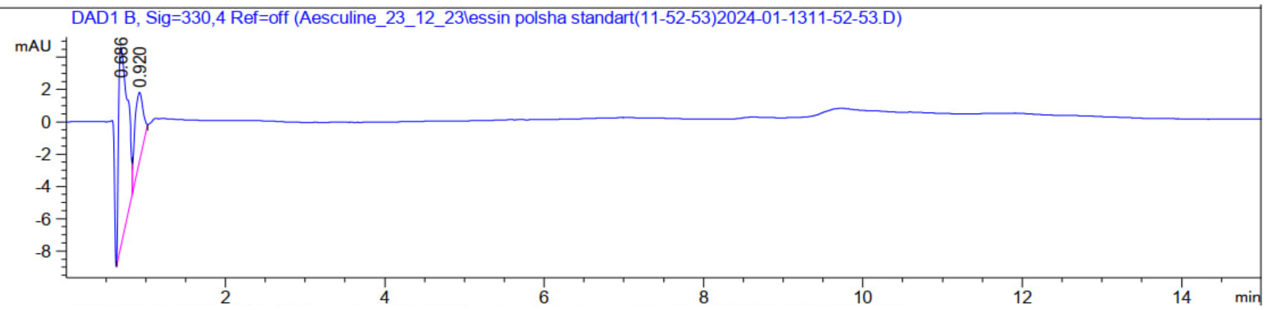


**Figure 2. Chromatogram of the analysis of coumarins in Escin substance.**

**Results.** Analysis of coumarins in various organs of horse chestnut grown in Uzbekistan showed that it is practically undetectable in organs other than its seeds and bark. In particular, the organ of horse chestnut that stores the most coumarins is its bark, containing 1.35% coumarins. It has been noted that horse chestnut seeds contain 1.7 times less (0.79%) coumarins than the bark. These indicators were compared with horse chestnuts growing in Europe, the homeland of horse chestnut, according to literature data. According to the data, 1.61% of coumarins are stored in European chestnut bark and 1.05% in chestnut seeds [Stanić G., Juričić B., Brkić D. HPLC analysis of esculin and fraxin in horse-chestnut bark (Aesculus hippocastanum L.) //Croatica chemica acta. – 1999. – T. 72. – №. 4. – C. 827-834.]. From this it is clear that horse chestnut, growing in its homeland, contains more coumarins. Coumarins are necessary to protect plant organs. It appears that the long-lived chestnut tree accumulates large amounts of coumarins to protect its most important organ, the bark. The fact that animals do not eat horse chestnut seeds can be explained by the coumarins they contain. The primary raw materials for obtaining the Escin substance are horse chestnut seeds; the amount of coumarins in them is 0.79%. The amount of coumarins in Escin substance was found to be 0.4±0.02%. Escin substance [FAP escin] is extracted from seeds using a patented technology. It consists of two stages; at the first stage, a dry extract is obtained from the seeds of horse chestnut. The amount of coumarins in it was analyzed and found to be 3.715%. It should be noted that the amount of escin in this dry extract is up to 20%. Then, "Escin substance" is obtained by deep recleaning of seeds. At the same time, the amount of coumarins decreases from 6.36% to 0.4%. The amount of escin exceeds 95%. It can be seen that the method presented in the article allows controlling the amount of foreign substances - coumarins in the full cycle of the production of "Escin substance" from horse chestnut seeds, which are primary raw materials. Thus, it is proposed to include the method for analyzing coumarins as a specific foreign substance into the regulatory documents of medicines obtained from horse chestnut. This, in turn, will be an excellent incentive to improve the quality of drugs containing escin.

**Conclusion.** A quantitative analysis of coumarins contained in various organs of horse chestnut growing in Uzbekistan was carried out for the first time and their largest amount was experimentally found in horse chestnut bark. A step-by-step quality control has been developed with the analysis of coumarins at all stages of the full cycle of the production of "Escin substance" from horse chestnut seeds, which are considered local raw materials. It is also proposed to include the developed method for analyzing coumarins in regulatory documents for medicines obtained from horse chestnut.

**References:**

[Stanić G., Juričić B., Brkić D. HPLC analysis of esculin and fraxin in horse-chestnut bark (Aesculus hippocastanum L.) //Croatica chemica acta. – 1999. – T. 72. – №. 4. – C. 827-834.].